MODULE -05

Mental Health Sentiment Analysis: Write journal entries, run them through an online sentiment analysis tool, and reflect on accuracy.

ABSTRACT

Mental health is a crucial component of human well-being, affecting emotional, psychological, and social functioning. With advancements in Artificial Intelligence (AI) and Natural Language Processing (NLP), sentiment analysis has become an effective technique for interpreting emotions expressed in textual data. This report presents a study on mental health sentiment analysis using self-written journal entries.

The study involves creating journal entries representing different emotional states such as happiness, sadness, anxiety, neutrality, and mixed emotions. These entries are analyzed using online sentiment analysis tools to evaluate how accurately the tools interpret human emotions. The results indicate that while sentiment analysis tools perform well in identifying clear positive and negative emotions, they struggle with complex emotional expressions such as anxiety and mixed feelings.

The report also discusses the methodology, results, limitations, applications, ethical concerns, and future scope of sentiment analysis in mental health monitoring. The findings highlight the importance of combining AI tools with human understanding for accurate mental health assessment.

1. INTRODUCTION

Mental health plays a vital role in determining how individuals handle stress, relate to others, and make decisions. In recent years, there has been increasing awareness about mental health issues such as stress, anxiety, and depression. However, identifying emotional states accurately remains a challenge.

With the growth of digital communication, people often express their emotions through written content such as:

- Personal journals
- Social media posts
- Blogs and online forums

This textual data provides an opportunity to analyze emotions using computational methods.

Sentiment analysis, also known as opinion mining, is a technique used to determine the emotional tone behind a body of text. It classifies text into categories such as positive, negative, or neutral. Advanced tools also attempt to detect emotions like joy, anger, fear, and sadness.

In the context of mental health, sentiment analysis can:

- Help identify emotional patterns
- Detect early signs of mental distress
- Provide insights into psychological well-being

This report focuses on analyzing personal journal entries using sentiment analysis tools and evaluating their effectiveness in capturing human emotions.

2. OBJECTIVES

The main objectives of this study are:

1. To understand the concept of sentiment analysis in mental health
2. To create journal entries representing various emotional states
3. To analyze these entries using an online sentiment analysis tool
4. To compare tool-generated results with actual emotions
5. To evaluate the accuracy and limitations of sentiment analysis
6. To explore real-world applications in mental health monitoring

3. LITERATURE REVIEW

Sentiment analysis has been widely used in areas such as marketing, customer feedback analysis, and social media monitoring. In recent years, researchers have explored its applications in mental health. Studies show that:

- Social media data can be used to detect depression and anxiety
- Machine learning models can identify emotional patterns in text
- NLP techniques can help in early intervention of mental health issues

However, research also highlights several challenges:

- Difficulty in understanding context
- Misinterpretation of sarcasm
- Limited ability to detect complex emotions

Despite these limitations, sentiment analysis remains a promising tool in mental health research.

4. METHODOLOGY

4.1 Overview

The methodology involves three main steps:

1. Writing journal entries
2. Analyzing them using a sentiment analysis tool
3. Evaluating the results

4.2 Data Collection (Journal Writing)

Five journal entries were created to represent different emotional states:

- Positive (Happiness)
- Negative (Sadness)
- Anxiety
- Neutral
- Mixed emotions

These entries simulate real-life personal reflections.

4.3 Journal Entries

Entry 1 – Happiness

"Today was a wonderful day. I felt confident and motivated. I completed my tasks successfully and spent quality time with friends. I feel truly happy and satisfied."

Entry 2 – Sadness

"I feel very low today. Nothing seems to work out for me. I keep thinking about my failures and it makes me feel hopeless."

Entry 3 – Anxiety

"I am feeling very nervous about my exams. My thoughts are racing and I can't concentrate properly. I feel restless and uneasy."

Entry 4 – Neutral

"Today was an ordinary day. I attended classes, completed assignments, and followed my routine. Nothing special happened."

Entry 5 – Mixed Emotions

"I achieved something important today, but I still feel stressed and uncertain about the future. I feel both happy and worried at the same time.

4.4 Tool Used

An online sentiment analysis tool such as:

- TextBlob
- VADER Sentiment Analyzer
- Google Natural Language API

was used to analyze the journal entries.

4.5 Procedure

1. Each journal entry was input into the tool

2. The tool analyzed the text
3. Output included:
    - Sentiment classification (Positive/Negative/Neutral)
    - Polarity score
4. Results were recorded and compared

5. RESULTS

Entry	Actual Emotion	Tool Result	Polarity	Accuracy
1	Happiness	Positive	High Positive	Accurate
2	Sadness	Negative	High Negative	Accurate
3	Anxiety	Negative	Moderate Negative	Partially Accurate
4	Neutral	Neutral	Near Zero	Accurate
5	Mixed	Slightly Positive	Low Positive	Inaccurate

6. ANALYSIS AND DISCUSSION

6.1 Performance Evaluation

The sentiment analysis tool performed well in identifying:

- Clearly positive emotions
- Clearly negative emotions
- Neutral statements

However, performance dropped when analyzing:

- Anxiety-related expressions
- Mixed emotional states

6.2 Key Observations

1. Accurate Detection of Simple Emotions
Entries with straightforward language (happy or sad) were classified correctly.

2. Misinterpretation of Anxiety
Anxiety is a complex emotion involving fear and uncertainty. The tool classified it simply as negative, ignoring its unique nature.

3. Failure in Mixed Emotions
In Entry 5, both positive and negative feelings were present. The tool detected only one dominant sentiment, leading to inaccurate results.

4. Dependence on Keywords
Words like “happy,” “sad,” and “stress” heavily influenced the results. The tool lacked deeper understanding.

7. LIMITATIONS

7.1 Lack of Context Awareness
The tool cannot fully understand the context of sentences.
7.2 Inability to Detect Complex Emotions
Emotions like:

- Anxiety
- Depression
- Emotional conflict
  are not clearly identified.

7.3 Ignoring Tone and Intent
Sarcasm and indirect expressions are often misinterpreted.

7.4 Limited Emotional Categories
Most tools classify text into only three categories:

- Positive
- Negative
- Neutral

8. APPLICATIONS IN MENTAL HEALTH

8.1 Early Detection
Helps identify signs of mental distress from written content.
8.2 Mood Tracking
Tracks emotional patterns over time using journals or social media.
8.3 AI-Based Therapy Tools
Used in chatbots for emotional support and guidance.
8.4 Social Media Monitoring
Detects at-risk individuals based on their posts.

9. ETHICAL CONSIDERATIONS

- **Privacy Protection:** Personal data must be kept confidential
- **User Consent:** Data should only be used with permission
- **No Replacement for Professionals:** AI cannot replace psychologists
- **Risk of Misinterpretation:** Incorrect results may cause harm

10. FUTURE SCOPE

Future improvements may include:

- Use of advanced AI models like deep learning
- Better understanding of context and tone

- Detection of complex emotions
- Integration with voice and facial analysis
- Personalized mental health monitoring systems

11. CONCLUSION

This study demonstrates that sentiment analysis is a useful tool for understanding emotional patterns in text. It performs well for simple emotions but struggles with complex and mixed emotional states. While sentiment analysis has great potential in mental health applications, it should be used as a supportive tool rather than a replacement for human judgment. Future advancements in AI will likely improve its accuracy and reliability.